

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								
	1	2	3	4	5	6	7	8

1. Connector & input protection

Field connector, 2-stage bidirectional TVS surge protection, chassis isolation

File: connector.kicad_sch

2. Power

eFuse (reverse/OV/UV), 60V buck to 5V, LDO 3.3V

File: power.kicad_sch

3. MCU

STM32G0B1CCT3, SWD, reset, internal temperature sensor

File: mcu.kicad_sch

4. Measurement

Capacitive humidity sensor (PCAP04) and Pt1000 2-wire (ADS1220)

File: measurement.kicad_sch

5. Analog outputs

2x DAC8760 V/I output with TPS26611 miswiring protection

File: analog_out.kicad_sch

6. RS-485

THVD2450 +/-70V fault-protected transceiver

File: rs485.kicad_sch

DOTECH HMT500 – Oil Moisture Transmitter – Schematic v0.6

v0.5: J1 = JST GH 1.25 mm 8P side entry (SM08B-GHS-TB), harness W-2 to M12 8P; J5 chassis spring contact (was M12 shell pin).

v0.4: J3 = JST SH 1.0 mm 4P (SM04B-SRSS-TB side entry, v0.5) for sensor harness W-1 direct from HTX99R (no feedthrough, mech Rev F); Pt1000 2-wire (Kelvin split at J3 pads).

v0.3: MCU STM32G0B1CCT3 (LQFP48, -40..125 C, in-house part); TPS26600PWP official pinout; single axial PCB 57x23 (mech Rev C).

Signal flow: J1 field connector -> protection -> eFuse -> 5 V buck -> 3.3 V LDO; sensor head J3 -> PCAP04 / ADS1220 -> MCU -> DAC8760 x2 / THVD2450 -> J1

Outputs: RS-485 Modbus RTU + 2x analog (4-20 mA / 0-20 mA / 0-10 V / 0-5 V selectable). Supply 12-30 V DC.

Protection target: any pin pair +/-30 V continuous (miswiring), surge +/-1 kV, ESD +/-8 kV contact.

Power symbols: GND, +3V3, +3V3A (analog 3.3 V), +5V, VIN_P (protected input), CHASSIS. Inter-sheet signals: global labels.

v0.6: parts cleanup (-16: pull-ups R11-15, LED D3/R16, RS-485 R60/61 D60/61, FB2 C19 C11 C2 C6); PCB temperature = ADS1220 + MCU internal sensors.

VERIFY: LMR36006, PCAP04, DAC8760, TPS26611 pin numbers are placeholders (datasheets not accessible when drawn). TPS2660: RTN link R22 to confirm.

TBD footprints: J5 chassis contact, CMC, GDT, buck inductor, LMR36006, PCAP04, DAC8760, TPS26611.

표기 규칙: 부품번호 = 굵은 글자 (R1, U4) / 부품값 = 보통 글자 (10k, DAC8760) / 전원 네트 = 기울임 글자 (+3V3, VIN_P)

신호 네트 = 테두리 있는 라벨 (SPL_SCK, OUT1_EXT) - 네트 이름은 부품번호·부품명과 겹치지 않게 지음

글자 겹침: 생성기가 부품 몸체·핀·선·라벨·메모와 겹치지 않는 자리에 자동 배치하고, 겹치면 생성 실패로 처리

VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22

Design notes: docs/hw/circuit-design.md

Oil moisture transmitter HMT500 – RS-485 + 2x V/I analog output

DOTECH Co., Ltd.

Sheet: /

File: HMT500.kicad_sch

Title: HMT500 Top

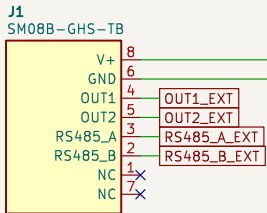
Size: A3Date: 2026-09-29Rev: 0.6

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4Id: 1/7

FIELD CONNECTOR

J1 JST GH 8P -> W-2 -> M12 8P

EE364 pinout, pin n = pin n

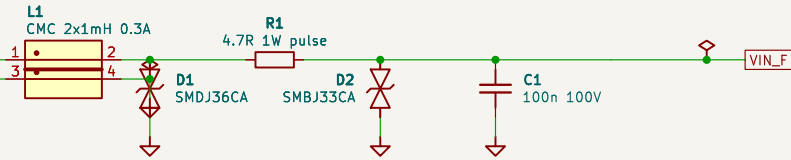


Pins 1, 7: not connected

GH mounting pads (MP): PCB only

INPUT SURGE / REVERSE-POLARITY PROTECTION

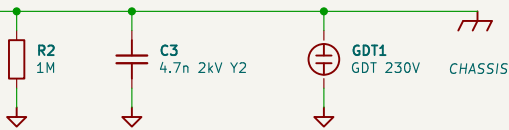
TVS bidirectional: -30 V miswiring must not conduct



SMDJ 3 kW (1st) -> R1 -> SMBJ (2nd) -> eFuse on sheet Power

CIRCUIT GND <--> CHASSIS (floating)

회로 GND-외함 분리



PCB-housing creepage >= 2 mm

GDT conducts only on line-to-ground surge

주요 부품 — 역할 · 특징 · 사양

부품번호	부품	역할	특징 · 사양
J1	SM08B-GHS-TB	현장 커넥터 연결	JST GH 1.25 mm 8P 옆 삽입, 하네스 W-2 -> M12 8P, 핀 n = M12 핀 n
L1	CMC 2x1mH 0.3A	전원선 공통모드 노이즈 차단	2 x 1 mH, 0.3 A — 전도 방출-전도 내성 대책
D1	SMDJ36CA	입력 서지 1단 흡수	TVS 양방향 3000 W, 36 V — -30 V 오결선에 도통 안 함
R1	4.7R 1W pulse	1단-2단 서지 분담	MELF 1 W 펄스 내량 저항
D2	SMBJ33CA	입력 서지 2단 클램프	TVS 양방향 600 W, 33 V — eFuse 정격 이하로 제한
GDT1	GDT 230V	회로 GND-외함 서지 방전	230 V 방전관, 선-대지 서지 때만 도통 (평소 절연)
C3, R2	4.7n Y2 / 1M	GND-외함 고주파 결합	Y2 안전 콘덴서 + 1 MΩ 정전기 방전 경로
J5	Chassis contact	PCB-하우징 접지 점검	SMD 스프링 점검, PCB 가장자리 -> 하우징 Ø27 내면

* 표시 값은 데이터시트로 확인

VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22
Design notes: docs/hw/circuit-design.md
Oil moisture transmitter HMT500 - RS-485 + 2x V/I analog output

DOTECH Co., Ltd.

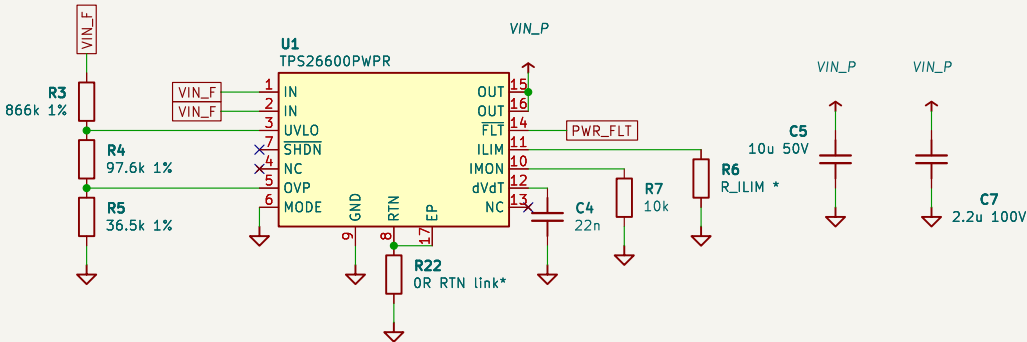
Sheet: /1. Connector & input protection/
File: connector.kicad_sch

Title: HMT500 Connector & input protection

Size: A3	Date: 2026-09-29	Rev: 0.6
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/7

eFuse TPS2660 reverse -60 V / OVP 33 V / UVLO 9 V

전자 퓨즈



$UVLO = 1.2\text{ V} \times (R3+R4+R5)/(R4+R5) = 8.9\text{ V}$

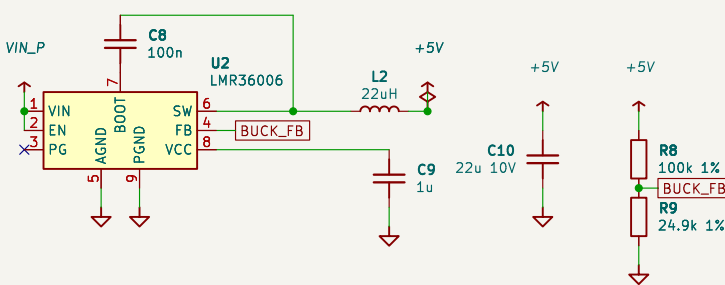
$OVP = 1.2\text{ V} \times (R3+R4+R5)/R5 = 32.9\text{ V}$ (Vref 1.2 V: VERIFY)

* R6 sets current limit 150 mA (formula in TPS2660 datasheet)

*R22 RTN link: verify vs TPS2660 datasheet – RTN tied to GND disables reverse–polarity protection

BUCK 5 V LMR36006 (4.2–60 V in, 0.6 A)

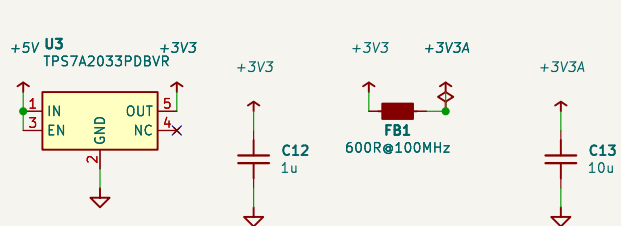
5 V 강압 전원



$V_{out} = 1.0\text{ V} \times (1 + R8/R9) = 5.0\text{ V}$ (Vref 1.0 V: VERIFY)

LDO 3.3 V TPS7A2033 + analog rail +3V3A

3.3 V 레귤레이터



주요 부품 — 역할·특징·사양

부품번호	부품	역할	특징·사양
U1	TPS26600PWR	전자 퓨즈 (입력 보호)	역극성 -60 V 차단, 과전압 33 V-저전압 9 V (R3-R5), 전류 제한 (R6)*, 돌입 제한 (C4), 고장 출력 FLT
U2	LMR36006	5 V 강압 전원 (벅)	입력 4.2-60 V, 출력 0.6 A 동기식, 무부하 소비전류 작음*, 출력 = 1 V × (1 + R8/R9)*
L2	22uH	벅 출력 인덕터	4 × 4 mm 차폐형, 포화 전류 0.8 A 이상으로 선정
U3	TPS7A2033PDBVR	3.3 V 저잡음 레귤레이터	300 mA, 저잡음·높은 PSRR* → 벅 리플 제거, SOT-23-5
FB1	600R@100MHz	아날로그 전원 +3V3A 분리	페라이트 비드 + C13 10 uF → PCAP04-ADS1220 전원

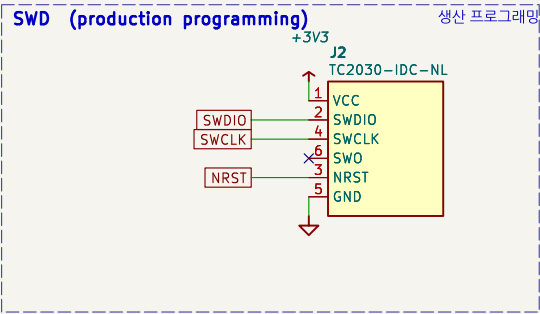
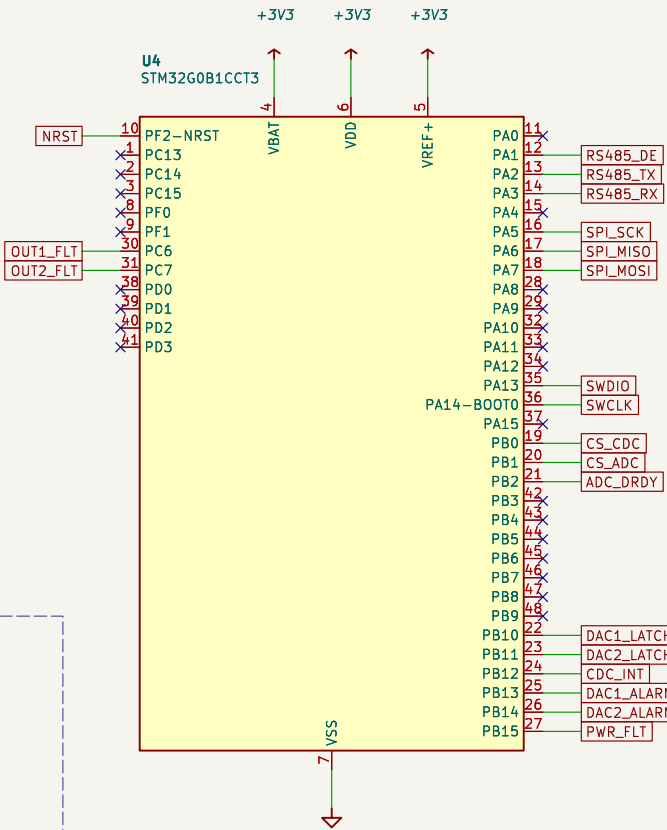
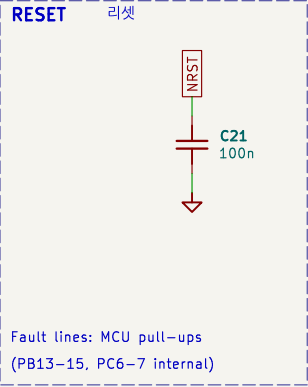
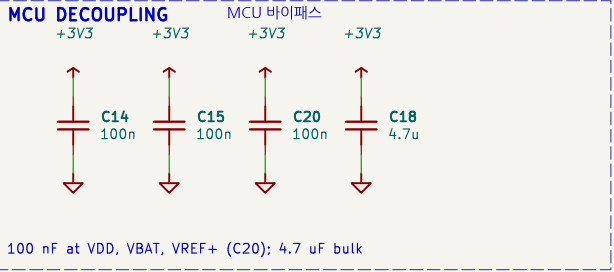
* 표시 값은 데이터시트로 확인

VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22
Design notes: docs/hw/circuit-design.md
Oil moisture transmitter HMT500 – RS-485 + 2x V/I analog output
NOTECH Co., Ltd.
Sheet: /2. Power/
File: power.kicad_sch

Title: HMT500 Power

Size: A3	Date: 2026-09-29	Rev: 0.6
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		

Id: 3/7



MCU NOTES (STM32G0B1) MCU 설정 메모

BOOT0 shares PA14/SWCLK - no pull-down.

Option bytes: nBOOT_SEL=1, nBOOT0=1

-> boot from flash (RS-485 bootloader 0x08000000).

VDDA = VDD (LQFP48); VREF+ tied to +3V3 (C20).

PCB temp (mid): internal sensor, ADC ch TS + TS_CAL.

Cortex-M0+: no SWO (J2 pin 6 NC).

Same die as DP2000 (G0B1CCT6); T3 = -40..125 C.

주요 부품 — 역할 · 특징 · 사양

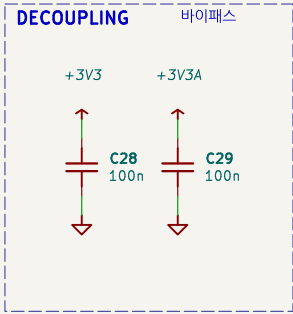
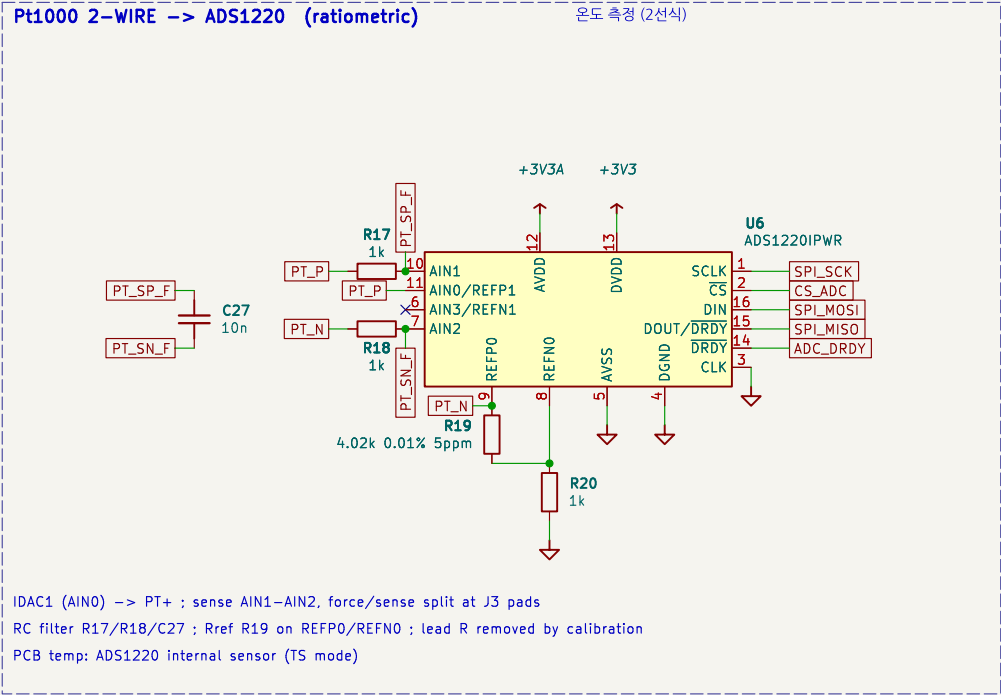
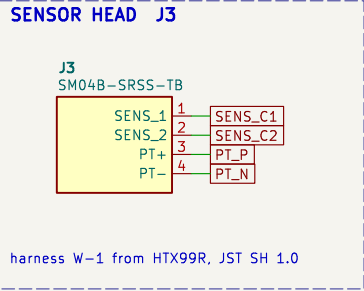
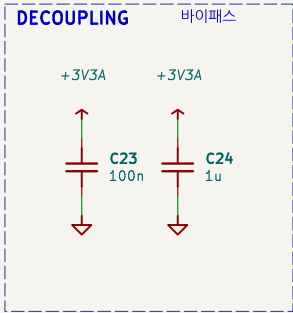
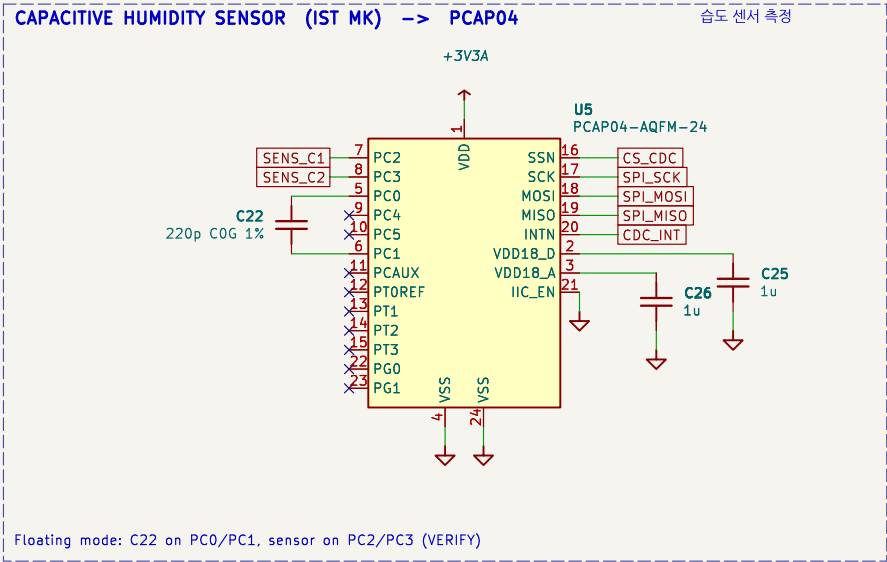
부품번호	부품	역할	특징 · 사양
U4	STM32G0B1CCT3	제어·통신·보정 연산	Cortex-M0+ 64 MHz, 플래시 256 KB, RAM 144 KB, -40-125 °C, LQFP48, 내부 온도센서 (PCB 온도)
J2	TC2030-IDC-NL	생산 때 펌웨어 기록 (SWD)	PCB 패드만 (부품 없음). 이후 업데이트는 RS-485 부트로더
C21	100n	리셋 노이즈 필터	NRST 핀, 물딩 후 EMC 여유

* 표시 값은 데이터시트로 확인

VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22
Design notes: docs/hw/circuit-design.md
Oil moisture transmitter HMT500 - RS-485 + 2x V/I analog output
DOTECH Co., Ltd.
Sheet: /3. MCU/
File: mcu.kicad_sch

Title: HMT500 MCU

Size: A3	Date: 2026-09-29	Rev: 0.6
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		
Id: 4/7		

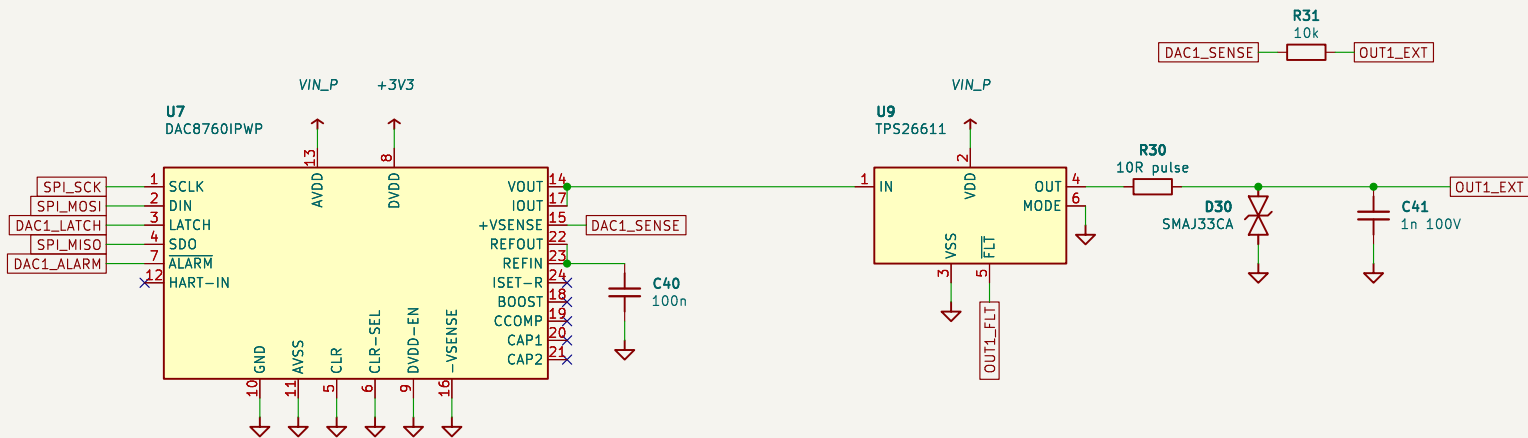


주요 부품 — 역할 · 특징 · 사양			
부품번호	부품	역할	특징 · 사양
J3	SM04B-SR55-TB	센서 하네스 W-1 연결	JST SH 1.0 mm 4P 옆 삽입, 1-2 = MK33, 3-4 = Pt1000
U5	PCAP04-AQFM-24	습도 센서 정전용량 측정	정전용량-디지털 변환, 기준 C와 비율 측정, 내부 DSP, SPI*
C22	220p COG 1%	PCAP04 기준 용량	MK33-W mini (200 pF)와 비슷한 값, COG (온도 변화 거의 없음)
U6	ADS1220IPWR	Pt1000 온도 측정	24비트 ADC, PGA, 여가 전류 IDAC, 50/60 Hz 제거, 내부 온도센서 (PCB 온도)
R19	4.02k 0.01% 5ppm	Pt1000 비율 측정 기준저항	IDAC 전류 오차 상쇄, 온도계수 5 ppm/°C
R17, R18, C27	1k / 10n	입력 RC 필터	차동-공통모드 노이즈 제거

* 표시 값은 데이터시트로 확인

ANALOG OUTPUT CH1 - V/I selectable, miswiring protected -> connector pin 4

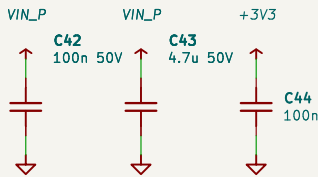
아날로그 출력 1



VOUT and IOUT combined (per TI TIDUBK2 - VERIFY). +VSENSE after R30 via R31.

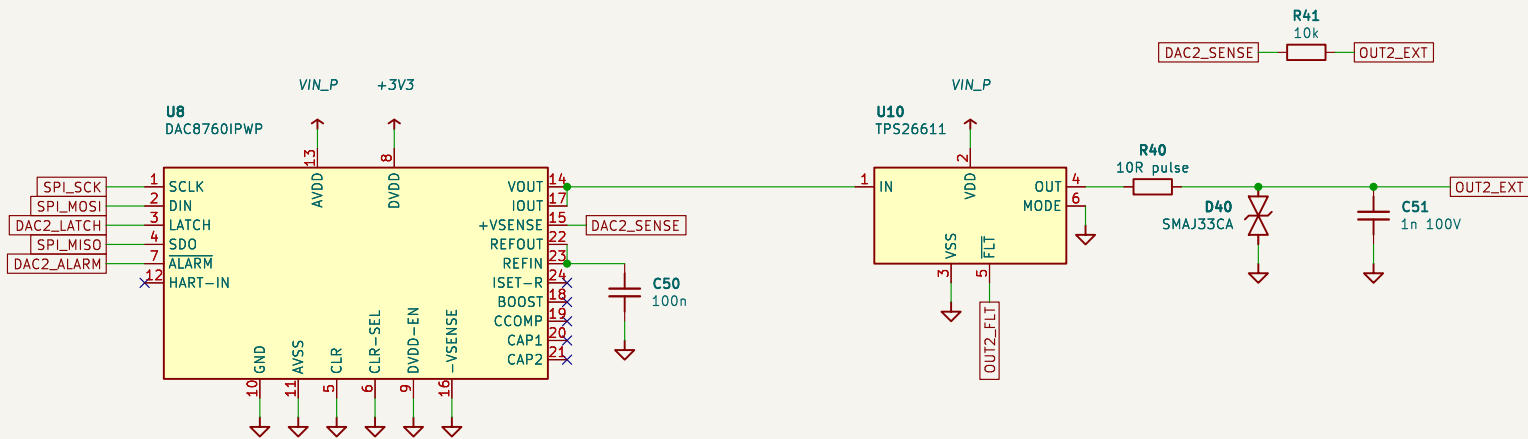
CH1 DECOUPLING

채널 1 바이패스



ANALOG OUTPUT CH2 - V/I selectable, miswiring protected -> connector pin 5

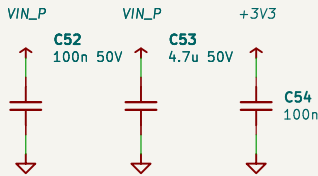
아날로그 출력 2



VOUT and IOUT combined (per TI TIDUBK2 - VERIFY). +VSENSE after R40 via R41.

CH2 DECOUPLING

채널 2 바이패스



DAC SELECTION (decision 2026-09-26: fit DAC8760 now; final part after output spec is fixed)

DAC 선정 (결정 #10)

U7/U8 fitted: DAC8760 (16-bit, current + voltage output). Same package/pinout family - drop-in alternates, no layout change:

DAC7760 12-bit, current + voltage (cost-down if V/I selectable output is kept)

DAC8750 16-bit, current only 4-20/0-20/0-24 mA (EE364-equivalent, 3-wire)

DAC7750 12-bit, current only (lowest cost; ~5 uA/step vs EE364 2 uA)

Current-only (x750): VOUT/+VSENSE/-VSENSE functions absent - check pin handling in datasheet before fitting (VERIFY).

주요 부품 - 역할 · 특징 · 사양

부품번호	부품	역할	특징 · 사양
U7, U8	DAC8760IPWP	아날로그 출력 1·2	16비트, 4-20 / 0-20 / 0-24 mA · 0-5 / 0-10 V 선택, 내부 기준, 고장 알람 - 대체 DAC7760/8750/7750
U9, U10	TPS26611	출력 오결선 보호	출력 단자 과전압·역전압 차단, 전류 제한, 고장 출력 FLT*
R30, R40	10R pulse (2512)	서지 전류 제한	출력 보호기와 TVS 사이 직렬
D30, D40	SMAJ33CA	출력 서지 클램프	TVS 양방향 400 W, 33 V
C41, C51	1n 100V	출력 고주파 필터	케이블로 들어오는 RF 억제
R31, R41	10k	전압 출력 감지	R30-R40 뒤에서 +VSENSE로 되먹임 (직렬 저항 전압강하 보상)

* 표시 값은 데이터시트로 확인

VERIFY=YES parts: pin numbers LMR36006/PCAP04/DAC8760/TPS26611; TPS2660 R22
Design notes: docs/hw/circuit-design.md
Oil moisture transmitter HMT500 - RS-485 + 2x V/I analog output

DOTECH Co., Ltd.

Sheet: /5. Analog outputs/
File: analog_out.kicad_sch

Title: HMT500 Analog outputs

Size: A3

Date: 2026-09-29

Rev: 0.6

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Id: 6/7

